PV101 Day 4 Morning Quiz - Answer Key

PV wiring, polarity, protection, and small-system sequencing

CTS Cohort 6 - PV101 | Day 4 | Thursday, July 9, 2026

Topic: Small Solar System Build: PV Panel, Charge Controller, 12.8V LiFePO4 Battery, Wiring, and Inverter

Class setup: 6 students; 5 kits. 5 kits for 6 students: run 5 build stations. One station is a two-student team with Builder/Inspector roles that swap halfway. Solo builders still require a neighbor peer-check before energizing.

Name: _____ Date: _____

Directions: Answer in plain field language. Complete sentences are welcome; correct thinking matters more than fancy vocabulary.

1. 1. What does voltage describe in a circuit?

Answer: Electrical pressure or potential difference.

2. 2. What does current describe in a circuit?

Answer: The flow of electric charge, measured in amps.

3. 3. In a PV circuit, why do we verify polarity before connecting equipment?

Answer: Reversed polarity can damage equipment and create unsafe conditions.

4. 4. What usually happens to voltage when two matching modules are wired in series?

Answer: Voltage adds; current stays about the same.

5. 5. What usually happens to current when two matching modules are wired in parallel?

Answer: Current adds; voltage stays about the same.

6. 6. Why should a fuse or breaker be close to the source of energy?

Answer: To protect conductors/equipment from overcurrent and reduce fault risk.

7. 7. Why does a charge controller need correct battery connection and polarity?

Answer: The controller must identify the battery/system voltage and reversed polarity can damage equipment.

8. 8. Why does an inverter usually connect to the battery side instead of the small controller load terminals?

Answer: An inverter can draw high current; it needs a properly protected battery connection unless the manufacturer says otherwise.

9. 9. What should be checked before uncovering or connecting the PV panel?

Answer: Circuit path, polarity, expected voltage, and instructor approval.

10. 10. Before energizing the small solar system, what should the group do?

Answer: Trace the circuit, verify polarity/voltage with a meter, check protection, and get instructor approval.